

भारतीय मानक
Indian Standard

IS 455 : 2015
(Reaffirmed 2020)

पोर्टलैंड धातुमल सीमेंट — विशिष्टि
(पांचवां पुनरीक्षण)

Portland Slag Cement —
Specification
(Fifth Revision)

ICS 91.100.10

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Price Group 5

Cement and Concrete Sectional Committee, CED 2

FOREWORD

This Indian Standard (Fifth Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Cement and Concrete Sectional Committee had been approved by the Civil Engineering Division Council.

This standard was first published in 1953 and subsequently revised in 1962, 1967, 1976 and 1989. This fifth revision incorporates the experience gained with the use of this specification and brings the standard in line with the latest developments in this field.

This standard pertains to Portland slag cement and covers the requirements such as its manufacture, chemical and physical requirements, packing and marking.

Since the last revision of this standard, a number of amendments were issued from time to time in order to modify various requirements based on experience gained with the use of the standard and the requirements of the users. These amendments have been incorporated in this revision so as to make it more convenient for the users. Further, following are the significant modifications incorporated in this revision:

- a) Reference to relevant Indian Standard has been given for the quality of clinker to be used for manufacture of Portland slag cement.
- b) SO₃ content requirement has been revised to 3.5 percent maximum, based on the studies in regard to the same, and also considering the use of coal/pet coke as fuel which may have higher sulphur content; subject to the cement conforming to all the requirements of the standard.
- c) A clause has been introduced requiring manufacturer to furnish a certificate indicating alkali content if required by the purchaser.
- d) Requirement of testing the cement samples at the earliest but not later than 3 months since the receipt of samples for testing, has been included.
- e) Requirement of marking the 'best before date' of cement has been introduced.

With the increase in SO₃ content limit in this revision, suitable caution needs to be exercised for limiting the sulphates in concrete in accordance with the provision of IS 456 : 2000 'Code of practice for plain and reinforced concrete (*fourth revision*)'.

Quantity of cement packed in bags and the tolerance requirements for the quantity of cement packed in bags shall be in accordance with the relevant provisions of the *Standards of Weights and Measures (Packaged Commodities) Rules*, 2011 and **B-1.2**. Any modification in these rules in respect of tolerance on quantity of cement would apply automatically to this standard.

This standard contains Table 2, SI No. (vii) and **13.2.1** which give an option to the purchaser and Table 3, SI No. (v) and **10.2, 10.3, 10.4** and **10.4.3**, which call for an agreement between the purchaser and the supplier.

The composition of the Committee responsible for the formulation of this standard is given in Annex C.

For the purpose of deciding whether a particular requirement of this standard is complied with, the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

Indian Standard

PORTLAND SLAG CEMENT — SPECIFICATION

(Fifth Revision)

1 SCOPE

This standard covers the manufacture and chemical and physical requirements of Portland slag cement.

2 REFERENCES

The standards given in Annex A contain provisions which, through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision and parties to agreements based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards indicated in Annex A.

3 TERMINOLOGY

For the purpose of this standard, the definition given in IS 4845 and the following shall apply.

3.1 Portland Slag Cement — An intimately interground mixture of Portland cement clinker and granulated slag with addition of gypsum and permitted additives or an intimate and uniform blend of ordinary Portland cement and finely ground granulated slag with addition of ground gypsum, if required.

3.2 Granulated Slag — Slag in granulated form is used for the manufacture of hydraulic cement. Slag is a non-metallic product consisting essentially of glass containing silicates and alumino-silicates of lime and other bases, as in the case of blastfurnace slag, which is developed simultaneously with iron in blastfurnace or electric pig iron furnace. Granulated slag is obtained by further processing the molten slag by rapidly chilling or quenching it with water or steam and air.

4 RAW MATERIAL

4.1 Granulated Slag

Granulated slag used in the manufacture of Portland slag cement shall conform to IS 12089.

4.2 Portland Cement Clinker

The Portland cement clinker used in the manufacture of Portland slag cement shall conform to IS 16353.

4.3 Portland Cements

Portland cement for blending with ground granulated slag shall conform to IS 269.

5 MANUFACTURE

Portland slag cement shall be manufactured either by intimately inter-grinding Portland cement clinker and granulated slag or by intimately and uniformly blending ordinary Portland cement and finely ground granulated slag, with required addition of gypsum/ground gypsum so as to produce a cement capable of complying with this standard. The granulated slag constituent shall not be less than 25 percent and not more than 70 percent by mass of Portland slag cement. The homogeneity of the mixture shall be guaranteed within ± 3 percent of the declared slag addition, in the same consignment.

When Portland slag cement is obtained by grinding granulated slag with Portland cement clinker, no material shall be added after burning, other than gypsum (natural mineral or chemical, *see* Note), water, and not more than a total of 1.0 percent of air-entraining agents or other agents including colouring agents, which have proved not to be harmful. The limitation of all such additions shall also apply to manufacture of Portland slag cement by blending process.

NOTE — Chemical gypsum shall be added provided that the performance requirements of the final product as specified in this standard are met with.

6 CHEMICAL REQUIREMENTS

When tested in accordance with the methods given in IS 4032, Portland slag cement, shall comply with the chemical requirements given in Table 1.

7 PHYSICAL REQUIREMENTS

Portland slag cement shall comply with the physical requirements given in Table 2.

8 STORAGE

The cement shall be stored in such a manner as to permit easy access for proper inspection and identification, and in a suitable weather-tight building to protect the cement from dampness and to minimize warehouse deterioration (*see also* IS 4082).

9 MANUFACTURER'S CERTIFICATE

9.1 The manufacturer shall satisfy himself that the cement conforms to the requirements of this standard and, if requested, shall furnish a test certificate to this

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Table 1 Chemical Requirements for Portland Slag Cement
(Clause 6)

Sl No. (1)	Characteristic (2)	Requirement (3)
i)	Insoluble residue, percent by mass, <i>Max</i>	4.0
ii)	Magnesia, percent by mass, <i>Max</i>	10.0
iii)	Total sulphur content calculated as sulphuric anhydride (SO ₃), percent by mass, <i>Max</i>	3.5
iv)	Sulphide sulphur (S), <i>Max</i>	1.5
v)	Loss on ignition, percent by mass, <i>Max</i>	5.0
vi)	Chloride content, percent by mass, <i>Max</i>	0.1 0.05 (for prestressed structures) See Note
vii)	Alkali content	

NOTE — Alkali aggregate reactions have been noticed in aggregates in some parts of the region. On large and important jobs where the concrete is likely to be exposed to humid atmosphere or wetting action, it is advisable that the aggregate be tested for alkali aggregate reaction. In the case of reactive aggregates, the use of cement with alkali content below 0.6 percent expressed as sodium oxide (Na₂O+ 0.658 K₂O), is recommended. Where, however, such cements are not available, use of alternative means may be resorted to for which a reference may be made to appropriate provisions on durability in the concrete codes. If so desired by the purchaser, the manufacturer shall carry out test for alkali content.

effect to the purchaser or his representative, within ten days of testing of the cement (except for 28 days compressive strength test results, which shall be furnished after completion of the test).

9.2 The manufacturer shall furnish a certificate indicating the alkali content, if requested.

10 PACKING

10.1 The cement shall be packed in any of the following bags:

- Jute sacking bag conforming to IS 2580,
- Multi-wall paper sacks conforming to IS 11761,
- Light weight jute conforming to IS 12154,
- HDPE/PP woven sacks conforming to IS 11652,
- Jute synthetic union bags conforming to IS 12174, or
- Any other approved composite bag.

Bags shall be in good condition at the time of inspection.

10.1.1 The net quantity of cement per bag shall be 50 kg subject to provisions and tolerance given in Annex B.

Table 2 Physical Requirements for Portland Slag Cement
(Clause 7)

Sl No.	Characteristic	Requirement	Method of Test, Ref to	
(1)	(2)	(3)	(4)	
i)	Fineness, m ² /kg, <i>Min</i>	225	IS 4031 (Part 2)	
ii)	Soundness:			
	a) By Le Chatelier method, mm, <i>Max</i>	10	} <i>see</i> Note 1	IS 4031 (Part 3)
	b) By autoclave test method, percent, <i>Max</i>	0.8		
iii)	Setting time:			
	a) Initial, min, <i>Min</i>	30	} <i>see</i> Note 2	IS 4031 (Part 5)
	b) Final, min, <i>Max</i>	600		
iv)	Compressive strength, MPa (<i>see</i> Note 4):			IS 4031 (Part 6)
	a) 72 ± 1 h, <i>Min</i>	16		
	b) 168 ± 2 h, <i>Min</i>	22		
	c) 672 ± 4 h, <i>Min</i>	33		
v)	Transverse strength (optional)	<i>See</i> Notes 3 and 4		IS 4031 (Part 8)

NOTES

1 In the event of cements failing to comply with any one or both the requirements of soundness specified in the above table, further tests in respect of each failure shall be made as described in IS 4031 (Part 3), from another portion of the same sample after aeration. The aeration shall be done by spreading out the sample to a depth of 75 mm at a relative humidity of 50 to 80 percent for a total period of 7 days. The expansion of cements so aerated shall be not more than 5 mm and 0.6 percent when tested by Le-Chatelier method and autoclave test respectively.

2 If cement exhibits false set, the ratio of final penetration measured after 5 min of completion of mixing period to the initial penetration measured exactly after 20 s of completion of mixing period, expressed as percent, shall be not less than 50. In the event of cement exhibiting false set, the initial and final setting time of cement when tested by the method described in IS 4031 (Part 5) after breaking the false set, shall conform to the value given in the above table.

3 By agreement between the purchaser and the manufacturer, transverse strength test of plastic mortar in accordance with the method described in IS 4031 (Part 8) may be specified. The permissible values of the transverse strength shall be mutually agreed to between the purchaser and the supplier at the time of placing the order.

4 Notwithstanding the compressive and transverse strength requirements specified as per the above table, the cement shall show a progressive increase in strength from the strength at 72 h.

10.2 The net quantity of cement per bag may also be 25 kg, 10 kg, 5 kg, 2 kg or 1 kg and packed in suitable bags as agreed to between the purchaser and the manufacturer but the bag shall be of the material and quality as given in with **10.1**. The quantity of cement in the bags shall also be subject to tolerances as given in Annex B for 50 kg bags.

10.3 Supplies of cement in drums or in bulk may be made by arrangement between the purchaser and the supplier (manufacturer or stockist).

NOTE — A single bag or container containing 1 000 kg and more net quantity of cement, shall be considered as the bulk supply of cement. Supplies of cement may also be made in intermediate bags/containers, for example, drums of 200 kg, by agreement between the purchaser and the manufacturer.

10.4 When cement is intended for export and if the purchaser so requires, packing of cement may be done in bags or in drums with net quantity of cement per bag or drum as agreed to between the purchaser and the manufacturer.

10.4.1 For this purpose, the permission of the certifying authority shall be obtained in advance for each export order.

10.4.2 The words 'FOR EXPORT' and the net quantity of cement per bag/drum shall be clearly marked in indelible ink on each bag/drum.

10.4.3 The packing material shall be as agreed to between the manufacturer and the purchaser.

10.4.4 The tolerance requirements for the quantity of cement packed in bags/drum shall be as given in **10.2.1** except the net quantity which shall be equal to or more than the quantity in **10.4**.

11 MARKING

11.1 Each bag or drum of cement shall be legibly and indelibly marked with the following:

- a) Manufacturer's name and his registered trade-mark, if any;
- b) The words 'Portland slag cement';
- c) Net quantity, in kg;
- d) The words 'Use no Hooks' on the bags;
- e) Batch/control unit number in terms of week, month and year of packing;
- f) Best before date (that is, 3 months from date of packing);
- g) The need for testing of cement more than 3 months old to check conformity before its use;
- h) Address of the manufacturer; and
- j) Percentage of slag addition.

11.2 Similar information shall be provided in the delivery advices accompanying the shipment of packed

or bulk cement and on cement drums (*see* **10.3**).

11.3 BIS Certification Marking

The cement may also be marked with the Standard Mark.

11.3.1 The use of the Standard Mark is governed by the provisions of the *Bureau of Indian Standards Act*, 1986 and the Rules and Regulations made thereunder. The details of conditions under which a license for the use of the Standard Mark may be granted to manufacturers or producers may be obtained from the Bureau of Indian Standards.

12 SAMPLING

12.1 A sample or samples for testing may be taken by the purchaser or his representative, or by any person appointed to superintend the work for the purpose of which the cement is required or by the latter's representative.

12.1.1 The samples shall be taken within three weeks of the delivery and all the tests shall be commenced within one week of sampling.

12.1.2 When it is not possible to test the samples within one week, the samples shall be packed and stored in air-tight containers and tested at the earliest but not later than 3 months since the receipt of samples for testing.

12.2 In addition to the requirements of **12.1**, the methods and procedure of sampling shall be in accordance with IS 3535.

12.3 The manufacturer or the supplier shall afford every facility, and shall provide all labour and materials for taking and packing the samples for testing the cement and for subsequent identification of cement sampled.

13 TESTS

13.1 The sample or samples of cement for test shall be taken as described in **12** and shall be tested in the manner described in the relevant clauses.

13.2 Independent Testing

13.2.1 If the purchaser or his representative requires independent tests, the samples shall be taken before or immediately after delivery at the option of the purchaser or his representative, and the tests shall be carried out in accordance with this standard on the written instructions of the purchaser or his representative.

13.2.2 The manufacturer/supplier shall supply, free of charge, the cement required for testing. Unless otherwise specified in the enquiry and order, the cost

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of the tests shall be borne as follows:

- a) By the manufacturer/supplier, if the results show that the cement does not comply with the requirements of this standard, and
- b) By the purchaser, if the results show that the cement complies with the requirement of this standard.

13.2.3 After a representative sample has been drawn, tests on the sample shall be carried out as expeditiously as possible (*see 12.1.1 and 12.1.2*).

14 REJECTION

14.1 Cement may be rejected, if it does not comply with any of the requirements of this specification.

14.2 Cement remaining in bulk storage at the factory, prior to shipment, for more than six months, or cement in bags, in local storage such as, in the hands of a vendor for more than 3 months after completion of tests, shall be retested before use and shall be rejected if it fails to conform to any of the requirements of this specification.

ANNEX A

(Clause 2)

LIST OF REFERRED INDIAN STANDARDS

<i>IS No.</i>	<i>Title</i>	<i>IS No.</i>	<i>Title</i>
269 : 2015	Specification for ordinary Portland cement (<i>fifth revision</i>)	4032 : 1985	Methods of chemical analysis of hydraulic cement (<i>first revision</i>)
2580 : 1995	Textiles — Jute sacking bags for packing cement — Specification (<i>third revision</i>)	4082 : 1996	Recommendations on stacking and storage of construction materials and components at site (<i>second revision</i>)
3535 : 1986	Methods of sampling hydraulic cements (<i>first revision</i>)	4845 : 1968	Definitions and terminology relating to hydraulic cement
4031	Methods of physical tests for hydraulic cement	4905 : 1968	Methods for random sampling
(Part 2) : 1999	Determination of fineness by specific surface by Blaine air permeability method (<i>second revision</i>)	11652 : 2000	Specification for high density polyethylene (HDPE)/polypropylene (PP) woven sacks for packing cement (<i>second revision</i>)
(Part 3) : 1988	Determination of soundness (<i>first revision</i>)	11761 : 1997	Specification for multi-wall paper sacks for cement (<i>first revision</i>)
(Part 5) : 1988	Determination of initial and final setting times (<i>first revision</i>)	12089 : 1987	Specification for granulated slag for manufacture of Portland slag cement
(Part 6) : 1988	Determination of compressive strength of hydraulic cement (other than masonry cement) (<i>first revision</i>)	12154 : 1987	Light weight jute bags for packing cement
(Part 8) : 1988	Determination of transverse strength and compressive strength of plastic mortar using prism (<i>first revision</i>)	12174 : 1987	Jute synthetic union bags for packing cement
		16353 : 2015	Specification for Portland cement clinker

ANNEX B

(Foreword and Clause 10.1.1)

TOLERANCE REQUIREMENTS FOR THE QUANTITY OF CEMENT PACKED IN BAGS

B-1 The average of the net quantity of cement packed in bags at the plant in a sample shall be equal to or more than 50 kg. The number of bags in a sample shall be as given below:

<i>Batch Size</i>	<i>Sample Size</i>
100 - 150	20
151 - 280	32
281 - 500	50
501 - 1 200	80
1 201 - 3 200	125
3 201 and over	200

The bags in a sample shall be selected at random. For methods of random sampling, IS 4905 may be referred to.

B-1.1 The number of bags in a sample showing a minus error greater than 2 percent of the specified net quantity (50 kg) shall be not more than 5 percent of the bags in the sample. Also the minus error in none of such bags in a sample shall exceed 4 percent of the specified net quantity of cement in the bag.

B-1.2 In case of a wagon/truck load of up to 25 tonne, the overall tolerance on net quantity of cement shall be 0 to 0.5 percent.

NOTE — The mass of a jute sacking bag to hold 50 kg of cement is 531 g, the mass of a 6-ply paper bag to hold 50 kg of cement is approximately 400 g, the mass of a light weight jute bag to hold 50 kg of cement is approximately 450 g, the mass of a HDPE/PP woven sack to hold 50 kg of cement is approximately 70 g/71 g respectively, and the mass of a jute synthetic union bag to hold 50 kg of cement is approximately 420 g.

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ANNEX C

(Foreword)

COMMITTEE COMPOSITION

Cement and Concrete Sectional Committee, CED 2

<i>Organization</i>	<i>Representative(s)</i>
Delhi Tourism and Transportation Development Corporation Ltd, New Delhi	SHRI JOSE KURIAN (Chairman)
ACC Ltd, Mumbai	SHRI S. A. KHADILKAR SHRI RAMAN SADANAND PARULEKAR (<i>Alternate</i>)
Ambuja Cements Limited, Mumbai	SHRI J. P. DESAI SHRI C. M. DORDI (<i>Alternate</i>)
Atomic Energy Regulatory Board, Mumbai	SHRI L. R. BISHNOI SHRI SAURAV ACHARYA (<i>Alternate</i>)
Builders' Association of India, Mumbai	REPRESENTATIVE
Building Materials and Technology Promotion Council, New Delhi	SHRI J. K. PRASAD SHRI C. N. JHA (<i>Alternate</i>)
Cement Manufacturers' Association, Noida	DR K. C. NARANG DR S. K. HANDOO (<i>Alternate</i>)
CSIR-Central Building Research Institute, Roorkee	SHRI S. K. SINGH SHRI SUBHASH GURRAM (<i>Alternate</i>)
CSIR-Central Road Research Institute, New Delhi	DR RAKESH KUMAR
CSIR-Structural Engineering Research Centre, Chennai	DR K. RAMANJANEYULU SHRI P. SRINIVASAN (<i>Alternate</i>)
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Indian Institute of Technology Roorkee, Roorkee	DR V. K. GUPTA DR BHUPINDER SINGH (<i>Alternate</i>)
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The India Cements Limited, Chennai	DR D. VENKATESWARAN SHRI S. GOPINATH (<i>Alternate</i>)
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Member Secretaries

SHRI SANJAY PANT
Scientist 'E' (Civ Engg), BIS
and
SHRI S. ARUN KUMAR
Scientist 'C' (Civ Engg), BIS

Cement, Pozzolana and Cement Additives Subcommittee, CED 2 : 1

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ACC Ltd, Mumbai	SHRI S. A. KHADILKAR SHRI RAMAN SADANAND PARULEKAR (<i>Alternate</i>)

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Institute for Solid Waste Research & Ecological Balance (INSWAREB), Visakhapatnam	DR N. BHANUMATHIDAS SHRI N. KALIDAS (<i>Alternate</i>)
Lafarge India Limited, Mumbai	SHRI M. K. CHAUDHARY SHRI YAGYESH KUMAR GUPTA (<i>Alternate</i>)
Maharashtra Engineering Research Institute, Nasik	SCIENTIFIC RESEARCH OFFICER ASSISTANT RESEARCH OFFICER (<i>Alternate</i>)
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National Hydroelectric Power Corporation Limited, Faridabad	SHRI A. K. JAIN
National Test House, Kolkata	SHRI D. V. S. PRASAD DR MOITRAYEE DEVI (<i>Alternate</i>)
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Ready Mixed Concrete Manufacturers' Association, Mumbai	SHRI VIJAYKUMAR R. KULKARNI

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The Ramco Cements Ltd, Chennai	SHRI BALAJI K. MOORTHY SHRI ANIL KUMAR PILLAI (<i>Alternate</i>)
Ultra Tech Cement Ltd, Mumbai	SHRI SUBRATO CHOWDHURY SHRI M. R. KALGAL (<i>Alternate</i>)
In personal capacity (<i>Type IV/17, President's Estate, New Delhi</i>)	SHRI K. H. BABU

Panel for Revision of Cement Standards, CED 2:1/P1

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AMENDMENT NO. 1 JUNE 2018
TO
IS 455 : 2015 PORTLAND SLAG CEMENT —
SPECIFICATION

(Fifth Revision)

(Page 2, Table 1, Note, third sentence) — Insert the following after the third sentence:

‘However, in the case of cement having a minimum slag content of 50 percent, a maximum alkali content of 0.9 percent, expressed as sodium oxide ($\text{Na}_2\text{O} + 0.658 \text{ K}_2\text{O}$) is recommended.’